D5 – Final Version 1.1

Functional Status

This document belongs to the unfolding layer of the set.

It shows how the structures described in D1–D3 manifest in real systems.

It contains:

- no new axioms,
- no new normative setting,
- no new operationalization.

Its function is:

to make the structural transferability of the solution space framework to different real domains visible.

1. Starting Point

The solution space describes:

- path-dependent development,
- structural opening and narrowing,
- reality coupling of selection.

These properties do not appear only in theoretical models, but in real systems:

- biological,
- technical,
- economic,
- institutional.

2. Selection Principle of the Examples

The following fields were not chosen to confirm the framework, but to show:

that the same structural logic is observable in different system types.

3. Fields of Application

A1 – Biological Systems (Brain and States of Consciousness)

Empirical studies show:

- in the waking state: high multi-scale complexity and integration
- under anesthesia or loss of consciousness: clear reduction of this structure

(Wang et al., 2025)

Interpretation in the solution space:

- functional complexity corresponds to stable developmental structures
- reduction of this complexity corresponds to structural narrowing

Core Point:

Development manifests as organization across multiple scales.

A2 – Technical Systems (AI and Adaptive Models)

Modern AI systems show:

- high performance under training conditions
- but partly deviating behavior under changed conditions

Observed phenomena:

- Alignment Faking
- Distribution Shift
- strategic pseudo-conformity

(Greenblatt et al., 2025/2026; Koondijk, 2025/2026)

Interpretation in the solution space:

- local behavior ≠ global path structure
- apparent stability can conceal structural deviation

Core Point:

Behavior alone is not sufficient for evaluating systems.

A3 – Platform Economy and Digital Systems

Digital platforms show:

- strong network effects
- lock-in structures
- path dependence of decisions

Interpretation in the solution space:

- Selective Funnelling → stabilization of dominant structures
- State-Space Expansion → new markets and function spaces

Core Point:

Markets do not develop only efficiently, but structurally.

A4 – Institutional Systems

Organizations show:

- stabilization of routines
- formalization of processes
- reduction of uncertainty

Interpretation in the solution space:

- stability can lead to structural narrowing
- formally correct processes can remain structurally problematic

Core Point:

Institutional stability is not the same as structural viability.

A5 – Antifragility and Disturbance

Some systems show:

- improvement under variability
- learning from disturbance
- non-linear adaptation

Other systems show:

- apparent stability
- but increasing fragility under stress

(Axenie et al., 2024)

Interpretation in the solution space:

- development encompasses not only stability
- but also the capacity for structural adaptation

Core Point:

Not every stability is robust.

A6 – Neuronal Communication and Resonance Coupling

Empirical and theoretical works on the Communication-through-Coherence hypothesis show that neuronal communication cannot be explained solely by anatomical connection. Functional

effectiveness arises where rhythmic synchronization, phase position, delay, and direction of communication coincide.
Interpretation in the solution space:
Connection only becomes solution-space-effective through appropriate coupling.
Core Point:
Not every connection is a relationship. Not every coupling is viable.
A7 – Markov Blankets and Biological Individuation
Works on Markov Blankets show that systems can be described through statistical boundaries that distinguish internal and external states and at the same time enable exchange via sensory and active states. Hierarchical self-organization can be understood as the formation of Markov Blankets of Markov Blankets.
Interpretation in the solution space:
Complexity does not arise through boundless networking, but through nested boundaries that enable exchange and preserve own structure.
Core Point:
Boundary is not the opposite of relationship, but its condition.
A8 – Information-Theoretic Self-Organization
Information-theoretic approaches distinguish between mere entropy reduction and genuine self-organization. What is decisive is whether interdependencies arise through the dynamics of the system, especially in redundant and synergistic forms.
Interpretation in the solution space:
Development is not valuable because possibilities are reduced, but when possibilities are condensed into viable relationships.
Core Point:
Organization is interdependency formation, not merely order.
A9 – Metastable Dynamics
Metastability research describes systems that remain mobile between integration and segregation. They are neither rigidly trapped in one state nor arbitrarily unstable.
Interpretation in the solution space:
Future openness requires binding that remains solvable.
Core Point:
Viable systems hold without imprisoning, and release without destroying.
A10 – Human–AI Collective Intelligence
Research on AI-enhanced Collective Intelligence shows that human–AI systems can be understood as complex, multi-layered networks. Their performance depends not only on individual agents, but on roles, trust, communication, task structure, diversity, and interaction quality.
Interpretation in the solution space:
AI expands the solution space only when it strengthens human and collective sense-making without replacing real judgment agency.
Core Point:

Collective intelligence is not an addition of performance, but quality of coupling.

4. Common Structure across All Fields

S1 – Difference between Behavior and Structure

Systems can appear locally stable while their global structure is unstable.

S2 – Path Dependence

Early decisions shape later possibilities and limit alternatives.

S3 – Invisible Risks

Relevant effects are often:

- not directly observable
- temporally delayed
- systemically distributed

S4 – Non-Linearity

Small changes can have large structural effects.

5. Significance for the Framework

These observations show:

The solution space does not describe a special domain,
but a structural form of development that repeats across domains.

6. Delimitation

This document shows:

- where the structure becomes visible

It does not show:

- how it is measured (→ D3)
- what is admissible (→ D2)
- how it is critiqued (→ D6)

7. Closing Sentence

The structures described in the solution space appear equally in biological, technical, and institutional systems.

They do not describe a single domain, but a recurring pattern of development.

Source Basis of this Document (updated)

- Wong, M. L., et al. (2023). PNAS
- Wong, M. L., Hazen, R. M., et al. (2025). Interface Focus
- Greenblatt, R., et al. (2025/2026). Alignment Faking in LLMs
- Koondijk, J. (2025/2026). Empirical Evidence for Alignment Faking in a Small LLM
- Wang, et al. (2025). Neural Complexity Across Brain States
- Axenie, C., et al. (2024). Antifragility in Complex Dynamical Systems